



Exploring the Atwood number impact on shock-driven hydrodynamic instability at pentagonal interface using discontinuous Galerkin simulations

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ABSTRACT

Investigation of Atwood number impact is crucial for comprehending the flow dynamics of the evolving hydrodynamic instability. In this paper, the Atwood number impacts on the shock-driven hydrodynamic instability at a backward-facing pentagonal interface in its early stages are explored numerically. The shock-driven pentagonal interface has a unique property in which the incident angle of the shock wave is constant along the interface edge. This property provides the ideal environment for the shock-refraction process. To examine the Atwood number impact, we consider five distinct gases, including sulfur hexafluoride, krypton, argon, neon, and helium, which are filled inside the pentagonal interface and surrounded by nitrogen gas. A mixed modal discontinuous Galerkin method is used to solve the two-dimensional Navier–Stokes–Fourier equations for two-component gas flows for numerical simulations. The simulation results illustrate that the Atwood number plays a crucial role in the flow fields of the shocked-pentagonal gas interface, which have not been reported in the literature. The flow fields are greatly impacted by the Atwood number, leading to complex wave patterns, shock focusing, jet formation, interface deformation, and vorticity generation. Further, the Atwood number effects on the flow fields are also thoroughly examined by a quantitative study based on the integral quantities and the interface features. Finally, a quantitative analysis is performed to examine the intrinsic differences between the evolution of the flow fields at the square and pentagonal interfaces.

1. Introduction

The shock-driven hydrodynamic instability, often referred as the Richtmyer–Meshkov instability (RMI), occurs when an interface between two fluids with different densities is abruptly driven by a planar shock wave [1,2]. Small disturbances gradually grow during the early stages of RMI development, leading to the emergence of “bubble” and “spike” structures [3]. The area where a lighter fluid enters a heavier fluid is known as the bubble, while the area where a heavier fluid enters a lighter fluid is known as the spike. The primary focus of RMI research is the mix zone width, which is defined as the distance between the outside edges of the bubble and the spike. In RMI, pressure gradient (∇p) and density gradient ($\nabla \rho$) are generally found to be misaligned, i.e. $\nabla p \times \nabla \rho \neq 0$. Consequently, a number of complex physical phenomena are introduced, including fluids mixing mechanism, wave patterns, and vorticity creation.

The RMI widely exists in variety of industrial and scientific applications, such as astrophysics [4,5], inertial confinement fusion (ICF) [6], combustion ramjet [7], and compressible turbulent flows [8]. For instance, in an ICF target, the RMI improves the mixing of hot fuel with colder material, lowering or even eliminating the thermonuclear yield. On the other hand, RMI improves the combustion efficiency of supersonic air-breathing engines by expanding the area where fuel and air mix. Furthermore, the remnant of the historical supernova of 1054, the Crab Nebula, has filament structures shaped by the RMI. Due to its numerous applications, it is imperative that the suppression or stimulation of mixing caused by the RMI be studied [9–12].

Shock-driven enclosed gas interfaces (i.e. shock-driven gas bubbles) have shown to be among the most important areas of study for understanding the physical mechanics of the RMI throughout time. The RMI

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research on shock-driven gas interfaces in a variety of configurations—non-polygonal (such as cylindrical, elliptical or spherical) and polygonal (such as triangular, square, rectangular or diamond) interfaces, for example—has been the subject of numerous investigations. These studies have been done from theoretical, experimental, and numerical perspectives. Haas and Sturtevant [13] experimentally explored the shock-driven cylindrical interface containing light and heavy gases to investigate the RMI flow structure. Quirk and Karni [14] corroborated the experimental results of Haas and Sturtevant using the numerical results of shock-driven cylindrical interfaces. Zabusky and Zeng [15] numerically examined a shocked heavy axisymmetric spherical interface and investigated the evolving upstream and downstream complicated wave patterns. Bagabir and Drikakis [16] examined numerically the effect of shock Mach numbers on the flow fields of the shock-driven cylindrical light interface. Ranjan and co-authors [17,18] investigated experimentally and numerically the effects of shock strengths and Atwood number on the flow development and time-dependent integral properties in a three-dimensional shock-driven spherical bubble. Shankar et al. [19] studied the non-negligible impacts of diffusive and viscous terms in shock-driven cylindrical bubble to study the mixing phenomena of the RMI flows.

Zhu et al. [20] conducted a computational investigation on the genesis and evolution of vortex rings, shock wave patterns, shock focusing processes, jet generation mechanisms, and interface deformation in the shock-driven light/heavy cylindrical bubbles. Ding et al. [21] studied the first contact effects experimentally and numerically on the shock-driven heavy cylindrical/spherical bubbles. Zhu et al. [22] clarified numerically the influences of different incident shock Mach numbers on the shock-driven heavy cylindrical bubble. Zhu et al. [23] presented a computational study on the evolution of a shocked SF_6 bubble with a transverse density gradient. Recently, Singh and colleagues [24, 25] studied numerically the behavior of shocked light/heavy cylindrical bubbles under thermal nonequilibrium circumstances of non-monatomic gases. Singh et al. [26] investigates numerically the flow physics of shock-driven elliptical interface at different gas densities. Further, Singh et al. [27] investigated the unfolding physical phenomena of RMI at SF_6 elliptical bubbles using numerical simulations, emphasizing the aspect ratio effects on the flow morphology.

Majority of earlier research have been on shock-driven non-polygonal gas interfaces in various configurations, with just a small amount of attention given to other interface types like polygonal. Shock-driven polygonal gas interfaces have recently been the subject of some research, providing favorable conditions for shock-refraction physical phenomena. Bates et al. [28] examined the growth of RMI both numerically and experimentally at the shock-driven heavy rectangular interface. Later, Luo and colleagues [29,30] investigated the initial interface impacts on the flow fields by means of an experimental investigation of the different shocked-light/heavy polygonal interfaces. Igra and Igra [31,32] studied the pressure distribution and intricate wave patterns within various shocked polygonal interfaces with different gases through a numerical research. Encouraged by these investigations, Singh [33] investigated numerically the impacts of Atwood number on the RMI development at square interfaces containing different gases. Subsequently, Singh and colleague [34,35] explored the shock Mach number effects on the RMI evolution at light/heavy gas square interfaces. Singh and co-authors [36,37] carried out a detailed numerical investigation to assess the thermal nonequilibrium impacts of non-monatomic gases on the flow dynamics of the shocked-light/heavy square interfaces. After then, Singh and co-authors [38,39] provided a comparative numerical study of the RMI growth between the shocked-square and rectangular light gas interfaces. Recently, Singh and Jalleli [40] investigated numerically the coupling effects on the RMI evolution at double heavy shock-driven square interfaces. Singh et al. [41] presented the flow simulations of shock-driven two square bubbles containing different gases.

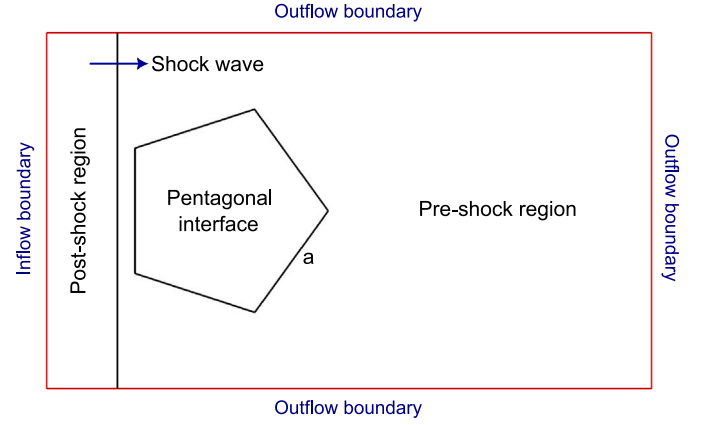


Fig. 1. Schematic description of the initial setup for a shock-driven backward-facing pentagonal interface, where a denotes the edge length of the pentagonal interface.

Inspired by previous developments, this study aims to investigate the impact of Atwood numbers on shock-driven hydrodynamic instability at a pentagonal interface, particularly focusing on the backward-facing configuration in its early stages. *To the best knowledge of the authors, no attempt has been made in the past to investigate the Atwood number effects on the flow physics at a shock-driven backward-facing pentagonal interface.* Furthermore, the current work lays the groundwork for future experimental and numerical study interpretation and analysis by providing a structured framework for the first theoretical and computational investigation of the Atwood number effect on the RMI at a pentagonal interface. The main emphasis of this framework is placed on the interface evolution, wave patterns, jet formation, vorticity generation, integral diagnostics and interface features of the shocked-pentagonal interface. This study offers prospects for additional research and can be viewed as a supplement to the shock-driven hydrodynamic instability research.

To achieve the primary goals of this investigation, a compressible two-component model has been utilized to simulate the RMI flows. To capture small-scale flow structures, numerical simulations are performed with a high-order mixed modal discontinuous Galerkin method. We now outline the structure of this paper. Section 2 illustrates the problem definition and mathematical model to simulate the RMI flows. Section 3 describes the employed numerical solver. Few validation test cases are showcased to verify the accuracy and reliability of the developed numerical solver. Section 4 provides the numerical results and detailed discussion on the Atwood number effects on the shock-driven pentagonal interface. Finally, conclusions remarks and outlook of the present study are given in Section 5.

2. Problem definition and mathematical formulation

2.1. Problem definition

A schematic diagram shown in Fig. 1 is the initial setup of a shock-driven pentagonal interface, where a rectangular domain of size $200 \text{ mm} \times 100 \text{ mm}$ is employed for conducting the numerical simulations. A moving planar shock wave and a stationary backward-facing pentagonal interface are considered in this study. The planar shock wave with Mach number $M_s = 1.2$ travels from left to right in the x -direction. The pentagonal interface is positioned at $x = 5 \text{ mm}$ from the shock wave, which originates at $x = 20 \text{ mm}$ from the left boundary of the domain as its initial location. The edge length of the pentagonal interface is fixed as $a = 30 \text{ mm}$ in this setup. In the computational domain, the left boundary is set to be inflow, while the upper, bottom and right boundaries are set to be outflow. The primary focus of this study is to explore the Atwood number impact on the shock-driven

Table 1Gas parameters at initial pressure $P_0 = 101,325$ Pa, and temperature $T_0 = 293$ K.

Gas	Density ($\rho_b, \text{kg m}^{-3}$)	Heat ratio (γ)	Sound speed ($c, \text{m s}^{-1}$)	Atwood number (At)	Flow pattern
N ₂	1.25	1.40	352	Ambient	–
He	0.16	1.66	1007	–0.773	Slow-fast
Ne	0.80	1.03	452	–0.218	Slow-fast
Ar	1.45	1.66	319	0.074	Fast-slow
Kr	3.43	1.66	222	0.466	Fast-slow
SF ₆	6.03	1.09	135	0.667	Fast-slow

pentagonal interface. Typically, in the shock-driven interface flows, Atwood number (At) can be defined as $\text{At} = (\rho_i) - (\rho_o)/(\rho_i) + (\rho_o)$, where ρ_i , and ρ_o denote the densities of unshocked interface gas and enclosing unshocked ambient gas, respectively. Here, the ambient gas is considered as nitrogen (N₂) with initial pressure $P = 101,325$ Pa and room temperature $T = 298$ K around the pentagonal interface, while a light gas (He or Ne) or heavy gas (SF₆, Kr or Ar) is used as the testing gas within the inhomogeneity. The used gas parameters for the numerical simulations are illustrated in Table 1.

2.2. Governing equations for two-component gas flows

In RMI studies, three-dimensional simulations are more useful compared to two-dimensional simulations because they offer a more realistic representation of the phenomenon, capture complex flow physics, provide predictive capability for practical applications, enable exploration of parameter space, and facilitate validation of experimental observations. However, they come with increased computational costs and challenges in data analysis and visualization. To consider these points, a two-dimensional system of unsteady compressible Navier–Stokes–Fourier (NSF) equations are utilized to conduct numerical simulations of the ideal non-reactive multicomponent flows [42]. The compressible NSF equations for an ideal gas mixture can be written in the conservative form as

$$\frac{\partial \mathbf{U}}{\partial t} + \nabla \cdot \mathbf{F}_{\text{inv}}(\mathbf{U}) + \nabla \cdot \mathbf{F}_{\text{vis}}(\mathbf{U}, \Pi, \mathbf{Q}) = 0, \quad (1)$$

where the conservation vector $\mathbf{U}$, the inviscid flux vector $\mathbf{F}_{\text{inv}}$, and the viscous flux vector $\mathbf{F}_{\text{vis}}$ are defined as

$$\mathbf{U} = \begin{bmatrix} \rho \\ \rho \mathbf{u} \\ \rho E \\ \rho Y_k \end{bmatrix}, \quad \mathbf{F}_{\text{inv}} = \begin{bmatrix} \rho \mathbf{u} \\ \rho \mathbf{u} \mathbf{u} + p \mathbf{I} \\ (\rho E + p) \mathbf{u} \\ \rho Y_k \mathbf{u} \end{bmatrix}, \quad \mathbf{F}_{\text{vis}} = \begin{bmatrix} 0 \\ \Pi \\ \Pi \cdot \mathbf{u} + \mathbf{Q} \\ \rho D_k \nabla Y_k \end{bmatrix}. \quad (2)$$

Here, ρ denotes the mass density; $\mathbf{u}$ is the velocity vector; p is the pressure; $\mathbf{I}$ is the unit second rank tensor; E is the total energy density; Π stands for the shear stress tensor; $\mathbf{Q}$ denotes the heat flux vector; Y_k is the mass fraction of species k ; and D_k is the diffusion coefficient of species k . In the above expression, the shear stress tensor Π and the heat flux vector $\mathbf{Q}$ are defined as follows

$$\Pi = -2\mu \left[\frac{1}{2}(\nabla \mathbf{u} + \nabla \mathbf{u}^T) - \frac{1}{3} \mathbf{I} \text{tr}(\nabla \mathbf{u}) \right], \quad \mathbf{Q} = -\kappa \nabla T, \quad (3)$$

where μ and κ are the coefficients of dynamic (shear) viscosity and thermal conductivity, respectively.

In this work, the mass fraction $Y_k = \rho_k/\rho$ is used to evaluate the fluid mixture where for two-fluid flows, $Y_k \equiv Y_1 = 1 - Y_2$. The total energy density E can be obtained as

$$\rho E = \frac{p}{\bar{\gamma} - 1} + \frac{1}{2} \rho |\mathbf{u}|^2, \quad (4)$$

where $\bar{\gamma}$ is the ratio of specific heats for the mixture, calculated as

$$\bar{\gamma} = \frac{\bar{C}_p}{\bar{C}_v} = \frac{\sum_k z_k C_{p,k}}{\sum_k z_k C_{v,k}}. \quad (5)$$

Here, $z_k = Y_k/M_k$; M_k is the molar weight for species k ; $C_{p,k}$ and $C_{v,k}$ are the specific heat constants for species k , which are given by

$$C_{p,k} = \frac{\gamma_k R_u}{\gamma_k - 1}, \quad C_{v,k} = \frac{R_u}{\gamma_k - 1}, \quad (6)$$

where R_u is the universal gas constant. For the gas mixture, Dalton's law of partial pressures $p = \sum_k p_k$ is used with $p_k = \rho_k R_k T$. The mixture viscosity $\bar{\mu}$ and the mixture thermal conductivity $\bar{\kappa}$ is calculated as [43]

$$\bar{\mu} = \frac{\sum_k \mu_k Y_k / M_k^{1/2}}{\sum_k Y_k / M_k^{1/2}}, \quad \bar{\kappa} = \frac{\sum_k \kappa_k Y_k / M_k^{1/2}}{\sum_k Y_k / M_k^{1/2}}. \quad (7)$$

In each fluid, the viscosity is calculated from Chapman–Enskog laws [44]

$$\mu_k = 2.6693 \times 10^{-6} \frac{\sqrt{M_k T}}{\Omega_{\mu,k} \sigma_k^2}, \quad (8)$$

where σ_k is the collision diameter, and $\Omega_{\mu,k}$ is the collision integral calculated by

$$\Omega_{\mu,k} = A(T_k^*)^{-B} + C e^{-DT_k^*} + E e^{-FT_k^*} \quad (9)$$

with constants $A = 1.16145$, $B = 0.14874$, $C = 0.52487$, $D = 0.77320$, $E = 2.16178$, $F = 2.4378$ and $T_k^* = \frac{T}{(\epsilon/K_B)_k}$, where $(\epsilon/K_B)_k$ is the Lennard-Jones potential of a gas and K_B is the Boltzmann constant. The thermal conductivity of species k is given by

$$\kappa_k = C_{p,k} \frac{\mu_k}{Pr_k}, \quad (10)$$

where Pr_k is the species specific Prandtl number. The mass diffusivity coefficient of a binary mixture is given by

$$D_{ij} = \frac{0.0266}{\Omega_D} \frac{T^{3/2}}{p \sqrt{M_{ij}} \sigma_{ij}^2}. \quad (11)$$

The collision integral for diffusion is given by [19]

$$\Omega_D = A(T_{ij}^*)^{-B} + C e^{-DT_{ij}^*} + E e^{-FT_{ij}^*} + G e^{-HT_{ij}^*} \quad (12)$$

where $M_{ij} = 2M_i M_j / (M_i + M_j)$, $\sigma_{ij} = (\sigma_i + \sigma_j)/2$, and $T_{ij}^* = \sqrt{(\epsilon_i/K_B)(\epsilon_j/K_B)}$. Also, the constants are specified as $A = 1.06036$, $B = 0.1561$, $C = 0.19300$, $D = 0.47635$, $E = 1.03587$, $F = 1.52996$, $G = 1.76474$ and $H = 3.89411$.

2.3. Initial conditions for simulations

To initialize the numerical simulations, an ambient condition on the right-hand side of the IS wave is employed. The primitive variables are calculated on the left-hand side of the IS wave using the standard Rankine–Hugoniot conditions. The standard Rankine–Hugoniot conditions for primitive variable calculations are expressed as [38]

$$M_2^2 = \frac{1 + \left[\frac{(\gamma-1)}{2} \right] M_s^2}{\gamma M_s^2 - \frac{(\gamma-1)}{2}}, \quad \frac{p_2}{p_1} = \frac{1 + \gamma M_s^2}{1 + \gamma M_2^2}, \quad \frac{\rho_2}{\rho_1} = \frac{\gamma - 1 + (\gamma + 1) \frac{p_2}{p_1}}{\gamma + 1 + (\gamma - 1) \frac{p_2}{p_1}}, \quad (13)$$

where M_s is the shock Mach number, and subscripts 1 and 2 refer to the left- and right-hand sides of the IS wave, respectively.

3. Numerical method and validations

In this study, two-dimensional compressible NSF equations (1) for multicomponent flows is solved using a mixed modal discontinuous Galerkin method. To address the higher-order derivatives present in the viscous component, a new additional auxiliary variable vector Θ is taken into consideration in this scheme [45,46]. The auxiliary variable vector Θ can be defined as the derivative of either primitive or conservative variables. For applying the mixed DG formulation, the NSF equations (1) can be written as a coupled system for $\mathbf{U}$ and Θ as

$$\Theta - \nabla \mathbf{U} = 0,$$

$$\frac{\partial \mathbf{U}}{\partial t} + \nabla \cdot \mathbf{F}_{\text{inv}}(\mathbf{U}) + \nabla \cdot \mathbf{F}_{\text{vis}}(\mathbf{U}, \Theta) = 0. \quad (14)$$

3.1. Spatial and temporal discretizations in mixed DG framework

To discretize the mixed form of conservation laws (14), the computational domain $D \in \mathbb{R}^2$ is subdivided into non-overlapping square elements $E_1, \dots, E_M$ and each of these elements are mapped to the reference element $E_0 = [-1, 1]^2$. The bi-linear mappings $\bar{X}_m : E_0 \rightarrow E_m$ with $\bar{X}_m(\xi, \eta) = (x, y)^T$ are defined as

$$\bar{X}_m(\xi, \eta) = \frac{1}{4} [\bar{x}_1(1-\xi)(1-\eta) + \bar{x}_2(1-\xi)(1+\eta) + \bar{x}_3(1+\xi)(1+\eta) + \bar{x}_4(1+\xi)(1-\eta)], \quad (15)$$

for $m = 1, \dots, M$, where $\{\bar{x}_1, \bar{x}_2, \bar{x}_3, \bar{x}_4\}$ are the four corners of the element E_m . Due to the uniform Cartesian mesh, Jacobian of the adopted mappings is to be constant i.e.

$$J = \frac{1}{4} \Delta x \Delta y = \frac{\Delta x^2}{4}, \quad (16)$$

for equal element side lengths $\Delta x = \Delta x_m = \Delta y_m = \Delta x$, $m = 1, \dots, M$.

Further, we define the finite element space

$$V_h^l = \{v \in [L^2(D)]^5 : v|_{E_m} \in [P^l(E_m)]^5, m = 1, \dots, M\}, \quad (17)$$

where $P^l(E_m)$ denotes the set of all polynomials of degree at most l on E_m . Now, on every element, each conserved variable $U \in \mathbb{R}^5$ is approximated by a polynomial of degree l in each spatial direction on E_0 such as

$$U_h(x, y, t)|_{E_m} = U_h(\xi, \eta, t) \approx \sum_{i,j=0}^{N_l} U_{ij} \psi_{ij}(\xi, \eta), \quad (18)$$

$$\Theta_h(x, y, t)|_{E_m} = \Theta_h(\xi, \eta, t) \approx \sum_{i,j=0}^{N_l} \Theta_{ij} \psi_{ij}(\xi, \eta),$$

where $\Theta_{ij} \approx \Theta_h(x, y, t)|_{E_m}$ and $U_{ij} \approx U_h(x, y, t)|_{E_m}$ are the time-dependent modal degrees of freedom for Θ and U , respectively, in the considered element E_m . While, $N_l = (l+1)^2$ is the number of required basis functions for l -exact DG approximation. For this work, the interpolating orthogonal scaled Legendre basis functions are adopted, which are defined as

$$\psi_{ij}(\xi, \eta) = \frac{2^i(i!)^2}{(2i)!} P_i^{0,0}(\xi) \otimes \frac{2^j(j!)^2}{(2j)!} P_j^{0,0}(\eta), \quad -1 \leq \xi, \eta \leq 1, \quad i, j \geq 0, \quad (19)$$

where $P_i(\xi)$, $P_j(\eta)$ represent the Legendre polynomials.

Now we start the discretization process by establishing the weak form of Eq. (14). This form is obtained by multiplying Eq. (14) by a test function ψ_h , integrating over the domain E_m , and performing an integration by parts:

$$\begin{aligned} & \int_{E_m} \Theta_h \psi_h dV + \int_{E_m} \nabla \psi_h \cdot U_h dV - \int_{\partial E_m} \psi_h U_h \cdot \mathbf{n} d\Gamma = 0, \\ & \frac{\partial}{\partial t} \int_{E_m} U_h \psi_h dV - \int_{E_m} \nabla \psi_h \cdot \mathbf{F}_{\text{inv}}(U_h) dV + \int_{\partial E_m} \psi_h \mathbf{F}_{\text{inv}}(U_h) \cdot \mathbf{n} d\Gamma \\ & - \int_{E_m} \nabla \psi_h \cdot \mathbf{F}_{\text{vis}}(U_h, \Theta_h) dV + \int_{\partial E_m} \psi_h \mathbf{F}_{\text{vis}}(U_h, \Theta_h) \cdot \mathbf{n} d\Gamma = 0. \end{aligned} \quad (20)$$

Here, $\mathbf{n}$ is the outward unit normal vector; V and Γ are the volume and boundary of the rectangular element E_k , respectively. The numerical solutions U_h and Θ_h are discontinuous across element interfaces, therefore the interface fluxes are not uniquely defined. The flux functions $U_h \cdot \mathbf{n}$, $\mathbf{F}_{\text{inv}} \cdot \mathbf{n}$, and $\mathbf{F}_{\text{vis}} \cdot \mathbf{n}$ appearing in Eq. (20) are replaced by a single-valued function defined at the cell interfaces, so-called the numerical flux functions represented as $\mathbf{H}^{\text{aux}}$, $\mathbf{H}^{\text{inv}}$, and $\mathbf{H}^{\text{alt}}$, respectively. As a result, the aforementioned weak formulation can be expressed as follows:

$$\begin{aligned} & \int_{E_m} \Theta_h \psi_h dV + \int_{E_m} \nabla \psi_h \cdot U_h dV - \int_{\partial E_m} \psi_h \mathbf{H}^{\text{aux}} d\Gamma = 0, \\ & \frac{\partial}{\partial t} \int_{E_m} U_h \psi_h dV - \int_{E_m} \nabla \psi_h \cdot \mathbf{F}_{\text{inv}}(U_h) dV + \int_{\partial E_m} \psi_h \mathbf{H}^{\text{inv}} d\Gamma \\ & - \int_{E_m} \nabla \psi_h \cdot \mathbf{F}_{\text{vis}}(U_h, \Theta_h) dV + \int_{\partial E_m} \psi_h \mathbf{H}^{\text{alt}} d\Gamma = 0. \end{aligned} \quad (21)$$

All the integrals appearing in the weak formulation (21) are calculated using the Gauss–Legendre quadrature rule with a proper number of integration points consistent with the accuracy required. According to the analysis, if P^l methods are utilized, the quadrature rules for the boundaries and interior (volume) of the elements must be precise for polynomials of degree $2l+1$ and $2l$, respectively [46]. Furthermore, to mitigate nonphysical oscillations in the computational solutions, a high-order moment limiter introduced by Krivodonova is employed [47]. By assembling all the elemental contributions together, the mixed modal DG formulation leads to a semi-discrete system of ordinary differential equations in time, which is solved with an explicit third-order SSP Runge–Kutta scheme [48].

3.2. Numerical flux schemes

In this study, the numerical flux at the elemental interfaces for the inviscid term is calculated utilizing the HLLC Riemann solver, which is specifically designed for two-component flows [49].

$$\mathbf{H}^{\text{aux}} = \frac{1 + \text{sign}(S^*)}{2} \mathbf{F}_L^* + \frac{1 - \text{sign}(S^*)}{2} \mathbf{F}_R^*, \quad (22)$$

where $\mathbf{F}_L^*$ and $\mathbf{F}_R^*$ are the middle fluxes, and S^* denotes the middle-wave speed. They can be computed by the following relations:

$$\begin{aligned} \mathbf{F}_k^* &= \mathbf{F}_k + S_k(U_k^* - U_k), \quad k = L, R, \\ S^* &= \frac{P_R - P_L + \rho_L u_L(S_L - u_L) - \rho_R u_R(S_R - u_R)}{\rho_L(S_L - u_L) - \rho_R(S_R - u_R)}, \end{aligned} \quad (23)$$

with

$$U_k^* = \frac{S_k - u_k}{S_k - S^*} \begin{bmatrix} (\rho)_k \\ (\rho u)_k \\ (\rho v)_k \\ (\rho E)_k + (S^* - u_k) \left(\rho_k S^* + \frac{P_k}{(S_k - u_k)} \right) \\ (\rho \phi)_k \end{bmatrix}, \quad k = L, R. \quad (24)$$

Here, S_L and S_R represent the left-wave and right-wave speeds, respectively, which can be evaluated as

$$\begin{aligned} S_L &= \min(S_L^*, 0), \quad S_R = \max(S_R^*, 0), \\ S_L^* &= \min(\bar{u} - \bar{c}, u_L - c_L), \quad S_R^* = \max(\bar{u} + \bar{c}, u_R + c_R), \\ \bar{u} &= \frac{\sqrt{\rho_L} u_L + \sqrt{\rho_R} u_R}{\sqrt{\rho_L} + \sqrt{\rho_R}}, \\ \bar{c}^2 &= \frac{\sqrt{\rho_L} c_L^2 + \sqrt{\rho_R} c_R^2}{\sqrt{\rho_L} + \sqrt{\rho_R}} + \frac{1}{2} \left(\frac{\sqrt{\rho_L} \sqrt{\rho_R}}{(\sqrt{\rho_L} + \sqrt{\rho_R})^2} (u_R - u_L)^2 \right). \end{aligned} \quad (25)$$

Furthermore, the alternating scheme [48] is considered for the viscous and auxiliary numerical fluxes $\mathbf{H}^{\text{alt}}$ and $\mathbf{H}^{\text{aux}}$.

$$\begin{aligned} \mathbf{H}^{\text{aux}}(U_h^L, U_h^R) &= \frac{1}{2} [U_h^L + U_h^R] + C_{11} [U_h^L - U_h^R], \\ \mathbf{H}^{\text{alt}}(U_h^L, \Theta_h^L, U_h^R, \Theta_h^R) &= \frac{1}{2} [\mathbf{F}(U_h^L, \Theta_h^L) + \mathbf{F}(U_h^R, \Theta_h^R)] - C_{11} [U_h^L - U_h^R] \\ &+ C_{12} [\Theta_h^L - \Theta_h^R], \end{aligned} \quad (26)$$

The flux is referred to as alternating flux when $C_{11} = 0$ and $C_{12} = 1/2$ or $-1/2$. Interestingly, with $C_{11} = C_{12} = 0$, the described flux scheme becomes central flow.

3.3. Validations of numerical solver

The current numerical model and related computational solver were validated by simulating a scenario comprising the shock-driven square light interface. A comparison of flow fields and evolving interfaces evolution between the experimental work of Zhai et al. [29] and the corresponding present numerical simulations is illustrated in Fig. 2. In this validation, a stationary square N_2 gas interface surrounded

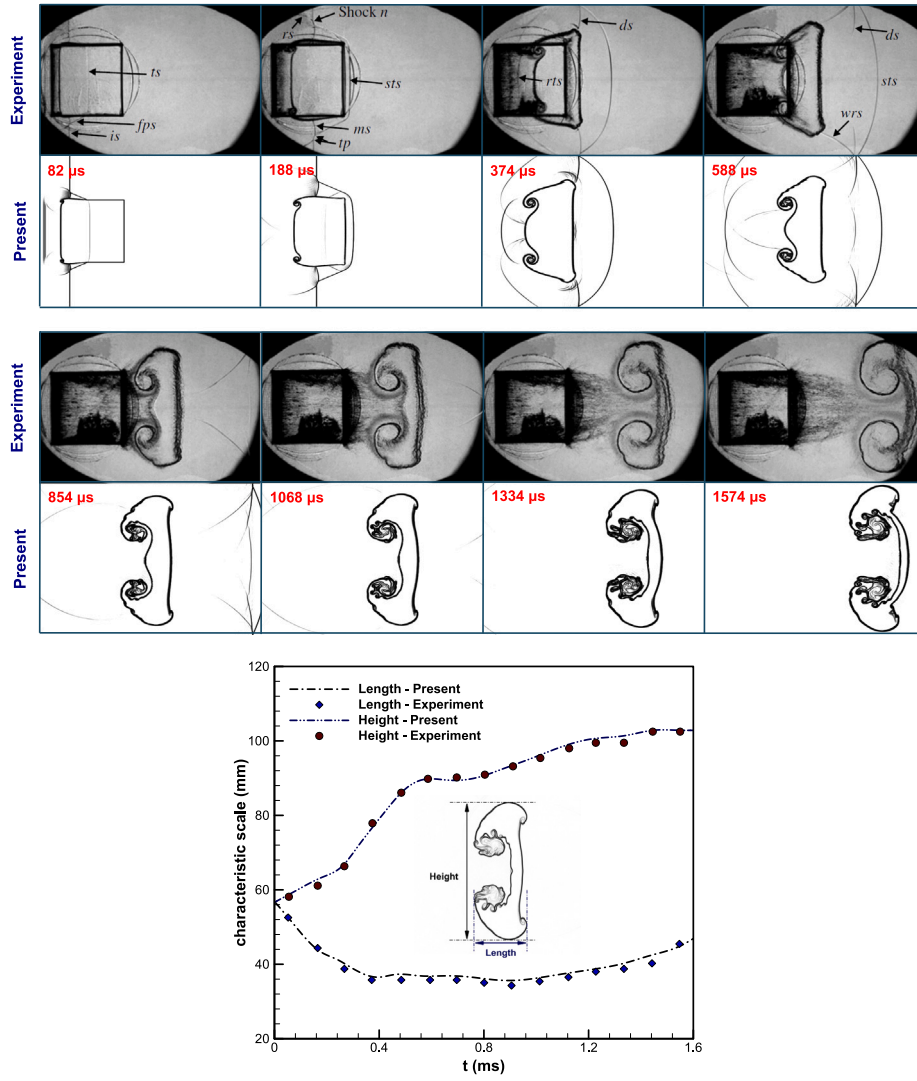


Fig. 2. Comparison of flow fields and evolving interfaces between the experimental work of Zhai et al. [29] and present numerical results for a shock-driven square light gas interface.

by a heavy SF_6 gas interacts with a weak moving shock of strength $M_s = 1.28$. Along with showing the identical interface evolution and intricate wave patterns seen by the experiment, the simulations also demonstrate how the propagating shock wave continuously deforms the square interface. Furthermore, Fig. 2 displays the time changes of the interfacial characteristic scales, i.e., the length and height of the evolving interface for the N_2 square bubble. The plot indicates that the current findings are found to be quite similar to the experiment conducted by Zhai et al. [29] including the general pattern of the interfacial characteristic scales changing with time. Thus, a high degree of agreement between the experimental and present results suggests the dependability of the current computational solver.

4. Results and discussion: Atwood number impact on shock-driven pentagonal interface

In this section, the Atwood number impact on the flow dynamics of a shock-driven regular backward-facing pentagonal interface in its early stages are comprehensively examined. Emphasis is placed on the Atwood number effect on the evolution of flow fields, wave patterns, vorticity generation, interface features, and various integral quantities. To this end, the following five tested gases listed in Table 1 are taken into consideration: helium (He), argon (Ar), sulfur hexafluoride (SF_6), krypton (Kr), neon (Ne), and argon (Ar). In contrast, the ambient gas

surrounding the pentagonal interface is thought to be nitrogen (N_2). All the numerical simulations are run with incident shock wave with $M_s = 1.2$.

In the following numerical simulations, a computational flow time is utilized to establish a dimensionless timescale defined as $\tau = t/t_0 = t W_i/a$, where $t_0 = a/W_i$ is characteristic time and W_i is the velocity of the shock wave. Furthermore, we use the numerical Schlieren images to visualize the computational results, which are based on the magnitude of density gradient i.e. $s = |\nabla \rho|$.

4.1. Grid resolution study

A grid resolution study is performed to precisely capture the complex flow field structure of the RMI flows. This entails calculating two test cases on the shock-driven pentagonal interface filled with SF_6 , and He gases. The grid study is performed on three grids: 400×200 , 800×400 , and 1200×600 with a third-order accurate modal DG solver. Fig. 3 shows the density distribution profiles obtained along the center-line of computed interface to further highlight the impact of grid resolution. The results reveal that the density dissipation reduces as the grid resolution increases. Moreover, we conducted enstrophy convergence analyses for the across five distinct grids: 400×200 , 800×400 , 1000×500 , 1200×600 , and 1400×700 , as illustrated

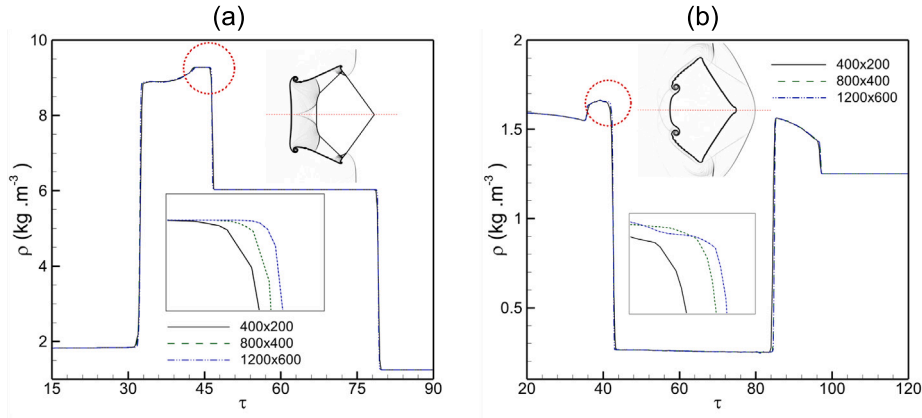


Fig. 3. Grid resolution study with three different grids in the shock-driven pentagonal interface filled with (a) SF_6 , and (b) He gases: density distribution profiles along the centerline of interfaces at $\tau = 10$.

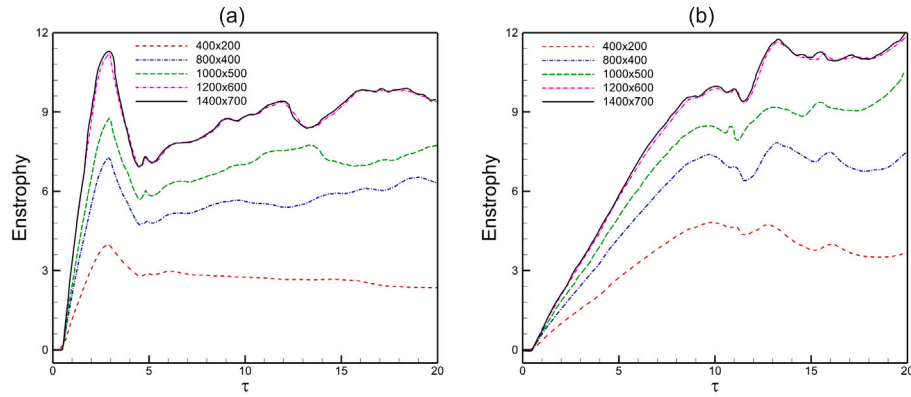


Fig. 4. Enstrophy convergence with five different grids in the shock-driven pentagonal interface filled with (a) He, and (b) SF_6 gases.

in 4. Through this comprehensive approach, we generated enstrophy evolution profiles, capturing the intricate dynamics of the system across varying resolutions. By systematically varying the grid resolution, we can assess the convergence behavior of enstrophy, a measure of vorticity dynamics, and its sensitivity to numerical discretization. The choice of grid resolution impacts not only the accuracy of our simulations but also the computational cost and resource requirements. Through our analysis, all the numerical computations are conducted using the 1200×600 grid points.

4.2. General flow configurations of a shock-driven pentagonal interface

Fig. 5 depicts a representative schematic view of the flow configurations in the shock-driven pentagonal interface. Typically, these arrangements display distinct flow phenomena and processes, like intricate wave patterns linked to the interaction and deformation of the pentagonal interface. As the incident shock wave strikes the leftmost surface of the pentagonal interface, alters the uniform front and divides into two portions. The transmitted shock wave is produced by the first half of the structure, whereas the second half does not directly affect the shock wave. Notably, the flow configuration is determined by the sign of At during the shock-interface interaction process. When $At < 0$ as shown in Fig. 4(a), the transmitted shock wave travels faster through the gas interface than the incident shock wave, resulting in a “slow-fast” or “light-heavy” configuration at the frontal region of the gas interface. In this scenario, the interface front attains a divergent shape. On the other hand, when $At > 0$, as shown in Fig. 4(b), this phenomenon is absolutely different. The transmitted shock wave travels through the gas interface more slowly than the incident shock wave, resulting in a “fast-slow” or “heavy-light” configuration at the

frontal region of the gas interface. The interface front thus achieves a convergent shape in this case.

4.3. Visualization of flow field evolution and wave patterns

Fig. 6 illustrates the time evolution of numerical Schlieren images of the shock-driven pentagonal He interface. In this case, the density of pentagonal He interface is lower than that of the surrounding N_2 , distinguished by a negative value of Atwood number ($At = -0.773$). This configuration means the shocked-pentagonal interface is in a “slow-fast” configuration. When the incident shock (IS) wave strikes the pentagonal interface, two waves are produced: a reflected shock (RS) that travels upward in the ambient and a direct transmitted shock (DTS) that propagates downward within the pentagonal ($\tau = 1$). This is recognized as the first reflection–transmission mechanism. Because of the low density inside the He interface, the DTS wave moves across it far more quickly than the IS wave from the outside. Simultaneously, the IS and DTS waves are connected by the free precursor shock (FPS) wave that is formed when the DTS wave refracts on the pentagonal interface once more. Consequently, an irregular pattern is developed, in which a Mach stem (MS), triple point (TP), reflected rarefaction shock (RRS), and a curved reflected shock (CRS) waves are involved ($\tau = 2$). Such an irregular refraction wave pattern is called “twin von Neumann refraction” (TNR) [50]. As time passes, the DTS strikes the downstream surface inside the pentagonal, resulting in the upward movement of the reflected free precursor shock (RFPs) and reflected transmitted shock (RTS) waves in the flow fields ($\tau = 2.5 - 3$). In addition, a second transmitted shock (STS) wave also observed outside the pentagonal interface. This might be thought of as the second transmission–reflection process. Due to the vorticity decomposition

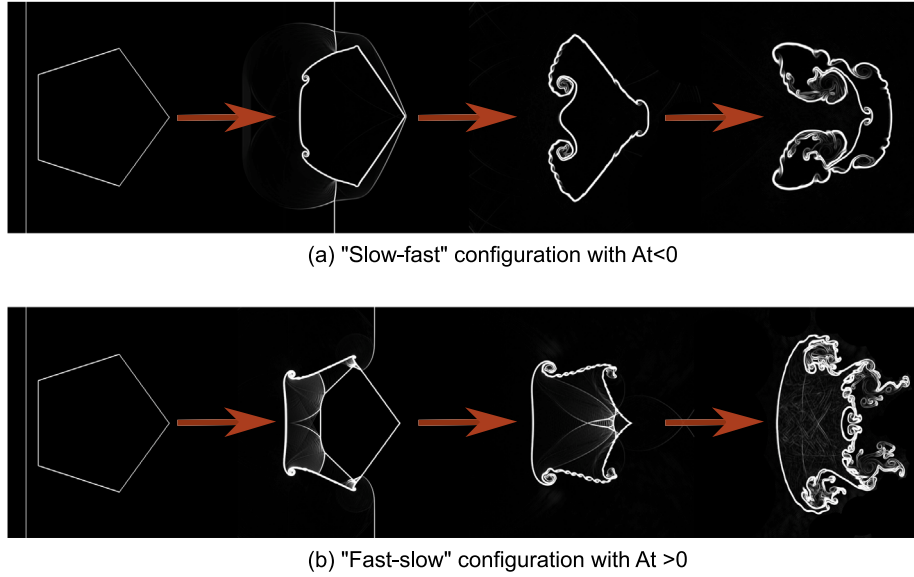


Fig. 5. Representative flow configurations of the shock-driven pentagonal interface: (a) "slow-fast" configuration with $At < 0$, and (b) "fast-slow" configuration with $At > 0$.

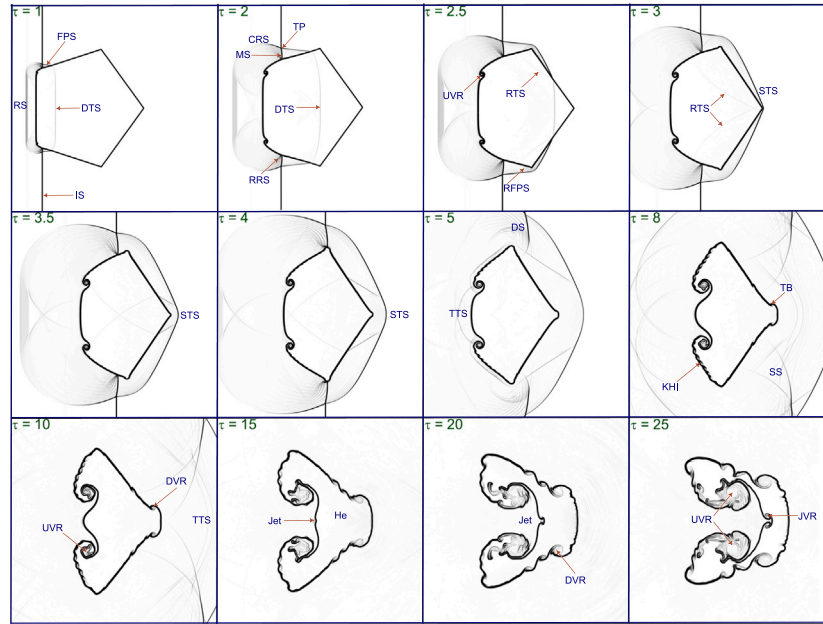


Fig. 6. Numerical Schlieren images for the shock-driven pentagonal He interface with $At = -0.773$ at different instants.

during the interaction, one pair of primary vortex rings (PVR) may be seen at the left corners of the interface. An ongoing interaction between the IS wave and pentagonal results in a diffracted shock (DS) wave along the rightmost interface. Furthermore, the RTS wave within the pentagonal travels upward before colliding with the upstream interface to produce a third transmitted shock (TTS) wave, also referred to as the third reflection-transmission process ($\tau = 5$). Meanwhile, more complex wave patterns, including slip surface, are experienced in the vicinity of the downstream interface. With time, the influence of the IS wave on the development of interface flow fields diminishes, and the vorticity production at the left corners causes the PVR to gradually grow. Moreover, the Kelvin-Helmholtz instability (KHI) causes some small-scale vortex formations to emerge on the leeward side contact ($\tau = 8$). Two trailing bubbles (TB) at the right corners of the pentagonal are also visible; these bubbles eventually intensify and develop into downstream vortex rings (DVR). Furthermore, a re-entrant helium jet is formed at the left interface of pentagonal, which starts to sink inward and

eventually takes on the shape of a mushroom ($\tau = 20$). A connecting bridge divides the produced bubble interface into symmetrical vortex rings (PVR, DVR) as the jet exits with the downstream interface. The PVR and DVR completely regulate the flow field and encourages the mixing enhancement between the interface and ambient gases ($\tau = 25$).

Fig. 7 illustrates the time evolution of numerical Schlieren images of the shock-driven pentagonal Ne interface with $At = -0.218$. The generated flow fields in the shock-driven pentagonal Ne interface are slightly different from those in the pentagonal He interface due to the slightly higher density of the Ne gas than He gas. A DTS wave travels inside the pentagonal at a slightly higher velocity than the IS when the IS wave interacts at the leading edge of the pentagonal interface. Simultaneously, it is noted that a RS wave is generated close to the left interface by the IS and pentagonal collision. As the interaction proceeds, the pentagonal Ne interface starts to compress, and both left-most interface corners slightly begin to bend due to the baroclinic vorticity ($\tau = 4$). The DTS generates RTS and STS waves into the

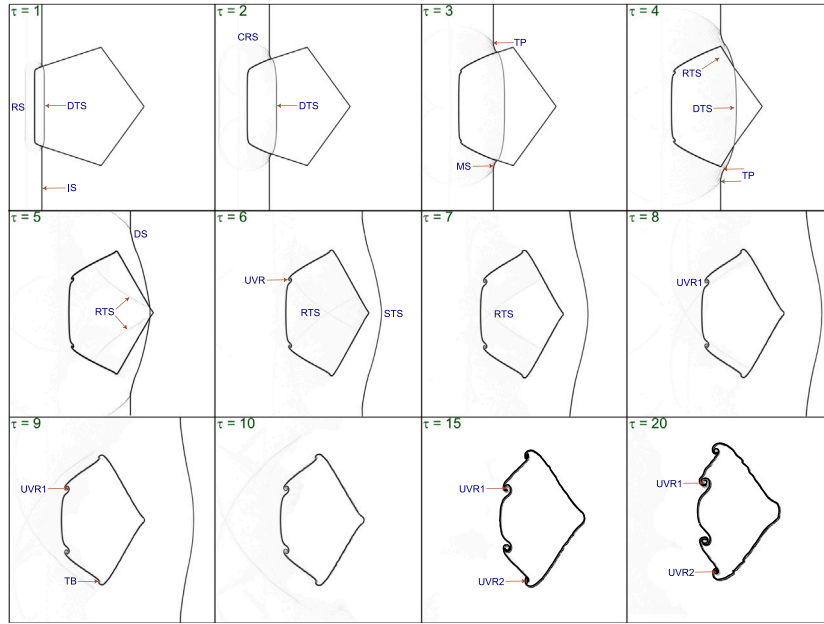


Fig. 7. Numerical Schlieren images for the shock-driven pentagonal Ne interface with $At = -0.218$ at different instants.

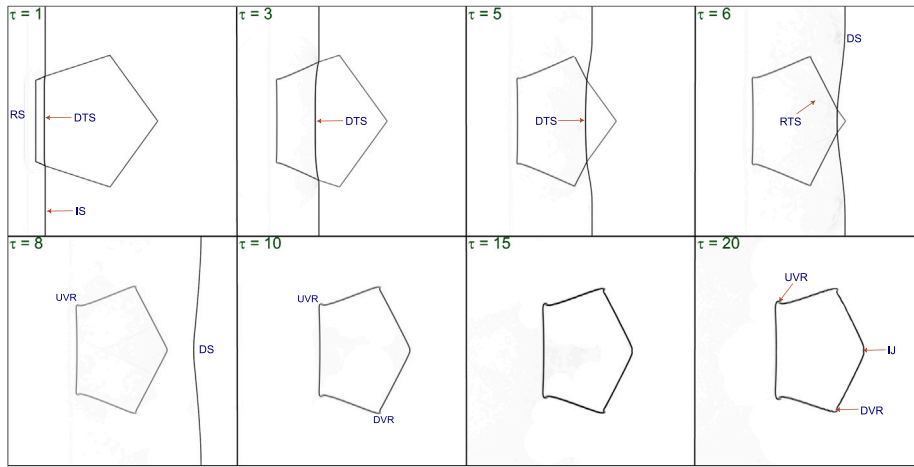


Fig. 8. Numerical Schlieren images for the shock-driven pentagonal Ar interface with $At = 0.074$ at different instants.

ambient N_2 gas as it strikes the rear surface of an interface, much like the shocked-pentagonal He interface. The RTS wave then leaves the pentagonal interface from its frontal surface and moves upstream into nitrogen gas, while the STS wave merges into a straight plane in front of the pentagonal. Interestingly, two upstream vortex rings (UVR1, UVR2) are observed in the frontal part of pentagonal interface ($\tau = 9$). The UVR1 and UVR2 are progressively more inwardly bent over time, and the interface's back corners are also gradually beginning to bow ($\tau = 10 - 20$).

Fig. 8 shows the time evolution of numerical Schlieren images of the shock-driven pentagonal Ar interface with $At = 0.074$. A positive Atwood number indicates that the density of Ar gas is marginally higher than the surrounding nitrogen when the density of the pentagonal gas contact keeps growing. This arrangement suggests that the shocked-pentagonal interface is in a “fast-slow” state. Due to the close resemblance in the acoustic impedance ($Z = \rho c$) of the relevant gases (argon and nitrogen), it is projected that the resulting wave pattern in this configuration would differ from the one seen with $At = -0.218$ and $At = -0.773$. As Fig. 8 illustrates, the velocities of the IS and DTS waves are in fact comparable. The propagation speed of the DTS wave inside the bubble is somewhat slower than that of the IS wave

outside during the initial stage of the IS wave interacting with the pentagonal interface. This is because the acoustic impedance of Ar gas is slightly greater than that of N_2 . Simultaneously, it is noted that an RS wave is generated near the left interface as a result of the interaction between the IS wave and the pentagonal. It is observed that the DTS into the Ar bubble reaches the pentagonal rear surface almost at the same time as the IS wave reached this location. Once the DTS shock wave within the pentagonal reaches the rear surface, the IS wave is transmitted into the ambient N_2 gas and RTS wave is generated inside the pentagonal. Later, this RTS wave breaks free from the pentagonal's frontal surface and moves upstream into the nitrogen gas. In comparison to the results obtained at $At = -0.218$ and for $At = -0.773$, as illustrated in Figs. 6–7, a different interface deformation is observed for $At = 0.074$. As a result of the contact, the frontal corners of the pentagonal interface gradually bend outward, generating upstream and downstream vortex rings. Furthermore, an inward jet IJ is observed at the center of downstream interface ($\tau = 20$).

Fig. 9 shows the time evolution of numerical Schlieren images of the shock-driven pentagonal Kr interface with $At = 0.466$. The greater acoustic impedance inside the Kr pentagonal interface causes the shock wave to propagate at a slower speed than it would in the

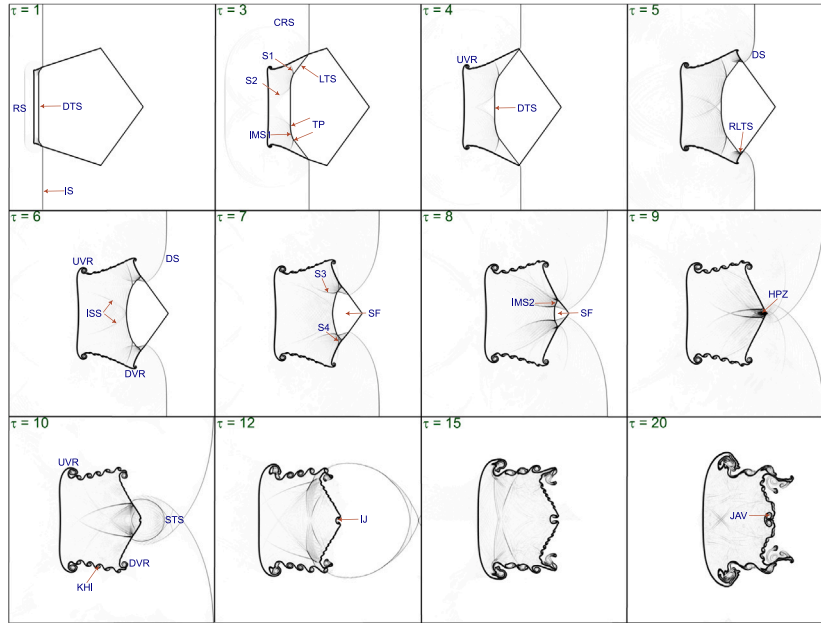


Fig. 9. Numerical Schlieren images for the shock-driven pentagonal Kr interface with $At = 0.466$ at different instants.

outside medium. Consequently, the transmitted shock wave inside the pentagonal is situated behind the planar incident shock wave outside the pentagonal. The frontal and lateral sides of the pentagonal are swept when the IS wave travels across the pentagonal ($\tau = 1 - 3$). After the IS wave impacts, a DTS wave is generated that appears to go downstream more slowly than the IS wave. The IS wave sweeps both the top and bottom of the lateral surfaces at a specific angle, creating a lateral transmitted shock (LTS) wave. It is interesting to observe the incident shock wave along the bubble interface (ISB), which moves more slowly than the IS wave. Besides it, a reflected shock (RS) wave traveling upstream simultaneously can also be observed. Two triple points (TP), an internal Mach stem (IMS1), and additional shocks (S1 and S2) are generated as a result of the irregular refraction that occurs when the IS wave propagates along the top and lower sides of the pentagonal ($\tau = 3$). In the meantime, a tiny UVR is formed by the depositions on the upper and lower interfaces and the baroclinic vorticity formation, which is the misalignment of the local pressure and density gradient. The top and bottom ends of the two straight parts of the IS wave are shown to be associated with a curving DS wave when the main portion of the wave moves ahead of the pentagonal interface ($\tau = 5$). Concurrently, an upstream-moving reversed lateral transmitted shock (RLTS) wave is detected inside the leeward corner of the pentagonal. With time, the phenomenon of uneven refraction also intensifies. As a result, two further shocks (S3, S4), an inward secondary shock (ISS) wave, and another internal Mach stem (IMS2) are detected in the interface downstream ($\tau = 6 - 8$). In addition, the collision of the DS and IS waves causes a noticeable shock focusing (SF) to occur inside the pentagonal around the leeward stagnation point, which create a high pressure zone (HPZ). The vortex pair UVR is continuously increasing in size over time and a small DVR is emerging ($\tau = 7$). Following focusing, one STS wave is produced and it grows downstream when the DS wave exits from the bubble downstream end ($\tau = 10$). Eventually, a bulge is observed at the gas interface. Soon afterward, as the bulge moves downstream, an inward Kr jet forms, driven by the HPZ formed by the intricate interplay of shocks within the pentagonal near the downstream centerline ($\tau = 12$). Moreover, rolled-up vortices become increasingly evident at the pentagonal interface as their small-scale size increases. The generated inward Kr jet expands along jet associated vortex (JAV) and vortex rings (UVR, DVR) over time.

Given that the gas density and acoustic impedance of SF_6 are both larger than those of Kr, as illustrated in Table 1, there should be some difference in the flow fields of the shock-driven pentagonal SF_6 interface. In order to examine these differences in more depth, the time evolution of numerical Schlieren images of the shock-driven pentagonal SF_6 interface with $At = 0.667$ are displayed in Fig. 10. Similar to shocked-pentagonal Kr interface, the RS wave can also be seen clearly in the frontal part of pentagonal, and the DTS, LTS, RLTS, DS and including other shock configurations can also be seen. The shock wave in the pentagonal SF_6 interface travels more slowly than in the Kr interface due to the larger acoustic impedance of SF_6 compared to Kr. As a result, the un-shocked region in the pentagonal SF_6 interface is larger than in the Kr interface. In addition, two slip lines—the upper slip line (USL), and the lower slip line (LSL)—inside the pentagonal is observed clearly ($\tau = 4$). A newborn transmitted shock (NTS), which has not been observed in prior cases, is an important observation to make during the shock focussing process. In contrast to the pentagonal Kr interface, the pentagonal SF_6 interface generates both UVR and DVR at its corners with greater intensities. Interestingly, a different pattern of inward jet formation is observed at the central part of the rear surface of the pentagonal SF_6 interface owing to the high pressure resulting from the shock focussing phenomenon ($\tau = 15$). Although there are some specific differences between the two different cases, due to the values of the Atwood number for the Kr and SF_6 gases, the overall trends of the Kr and SF_6 pentagonal in the evolution of shock waves and interface deformation are basically consistent.

Fig. 11 presents a comparative investigation of the Atwood number impact on the shock-driven pentagonal interface using numerical Schlieren pictures at the same early stages $\tau = 3$. The density and acoustic impedance are the factors that influence the deformation of the pentagonal interface for each Atwood number during the penetration of the IS wave. In the shocked-pentagonal He and Ne interfaces, the penetration speed are the faster, but these are also slower than in the IS wave, as shown in Figs. 11(a)–(b). In these situations, the DTS moves faster inside the pentagonal interface than the IS wave does outside of it. This situation is different in the case of shocked-pentagonal Ar interface, where a slight difference between the speeds of DTS and IS waves results in a smaller deformation of the interface, as illustrated in Fig. 11(c). The situation for the Kr and SF_6 gases is the opposite of what is seen in the cases of Ne and He gases, as shown in Figs. 11(d)–(e). The

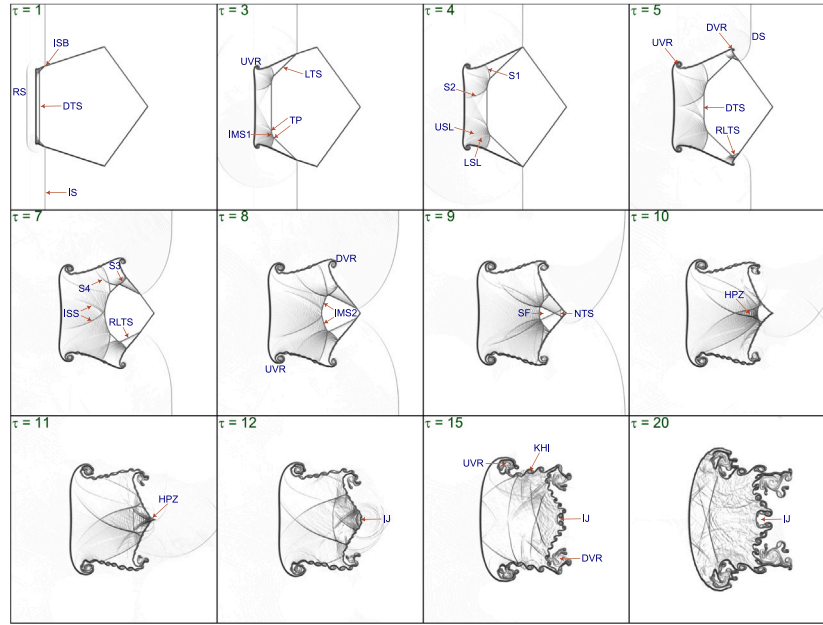


Fig. 10. Numerical Schlieren images for the shock-driven pentagonal SF_6 interface with $A_t = 0.667$ at different instants.

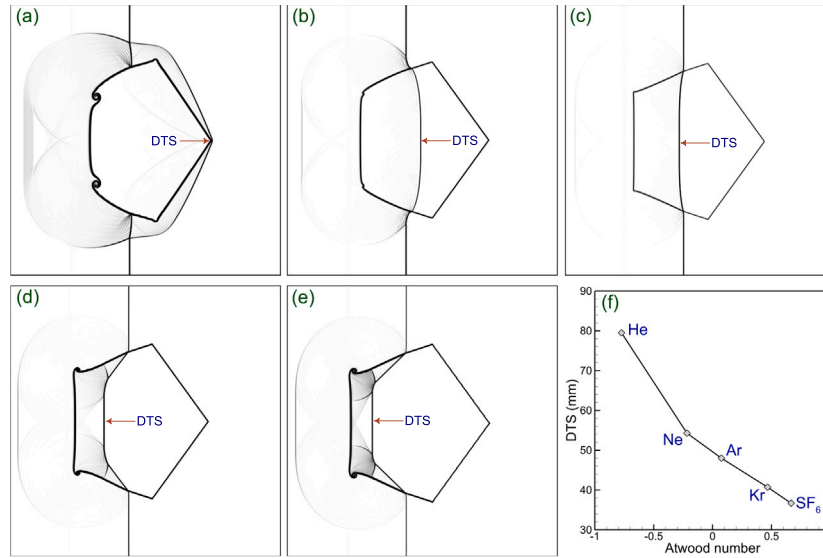


Fig. 11. Atwood number impact on the DTS displacement at $\tau = 3$: (a) $At = -0.773$, (b) $Ne = -0.218$, (c) $At = 0.074$, (d) $At = 0.466$, (e) $At = 0.667$, and (f) DTS position vs Atwood number.

slowest speed of penetration can be seen in SF_6 and Kr cases, and these speeds are at the same time slower than the IS wave. In these situations, the DTS moves more slowly inside the pentagonal interface than the IS wave does outside of it. The relationship between the Atwood number and the displacement of the DTS wave, as seen in Fig. 11(f), can be used to interpret the observation. It can be seen that the DTS wave moves through the bubble more slowly as the Atwood number increases and moves towards positive values.

4.4. Mechanism of vorticity generation

In shocked interface flows, vorticity generation mechanism involves complex interactions between shock waves, density gradients, shear flows, and other fluid properties. The resulting vorticity plays a crucial role in shaping the overall flow patterns and can have significant implications for various physical phenomena, such as vortex dynamics, mixing and turbulence. In such mechanism, the vorticity transport

equation (VTE) helps in understanding how vorticity is transported and influenced by the shock wave. For compressible viscous fluids, the VTE can be described as

$$\frac{D\omega}{Dt} = (\omega \cdot \nabla)\mathbf{u} - \omega(\nabla \cdot \mathbf{u}) + \frac{1}{\rho^2}(\nabla \rho \times \nabla p) + \frac{\mu}{\rho} \nabla^2 \omega, \quad (27)$$

On the right-hand side of above expression, the term $\omega(\nabla \cdot \mathbf{u})$ represents the fluid convection and thermal expansion which is only appropriate for highly compressible fluids. The term $(\omega \cdot \nabla)\mathbf{u}$ represents the vortex stretching, which is critical for the 3D turbulence and mixing discussion. The term $(1/\rho^2)(\nabla \rho \times \nabla p)$ denotes the baroclinic vorticity, and is responsible for the generation of small-scale rolled-up vortices at the gas interface. Finally, the term $(\mu/\rho)\nabla^2 \omega$ represents the acceleration due to the diffusion of vorticity.

Fig. 12 shows a schematic diagram of vorticity generation at the early stage of the shock-driven pentagonal interfaces in heavy ($At > 0$), and light ($At < 0$) gases. In shocked-interface flows, the interface is

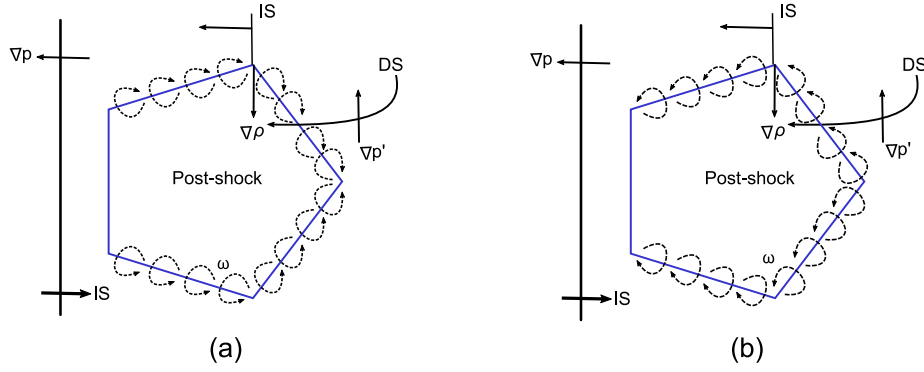


Fig. 12. Schematic diagrams of vorticity generation on the boundaries of pentagonal interface after initial shock wave transit in (a) heavy gas ($At > 0$), and (b) light gas ($At < 0$). The ∇p -pressure gradient caused by the incident shock wave; $\nabla p'$ -pressure gradient caused by the diffracted shock wave.

dominated by the density gradient, whereas the IS wave is dominated by the pressure gradient. When the IS wave passes through the pentagonal interface, the baroclinic vorticity is produced and distributed along the interface. The baroclinic vorticity is caused by the mismatch between the density and pressure gradients i.e. $\nabla \rho \times \nabla p \neq 0$, which has a large influence on the RMI development. After passing over the IS wave, the bubble does not deform significantly. However, some vorticity quantities in terms of small-scale vortices are produced at the leftmost vertical interface of the pentagonal, where the pressure and density gradients are perfectly aligned. The orientation of the density gradient at the pentagonal interface determines the rotation of these vortices. Moreover, a Mach reflection occurs as the IS wave communicates via the horizontal interfaces, with the Mach stem connecting the IS wave to the pentagonal interface. Consequently, the Mach stem also contributes to the pressure gradient in the vorticity generation, and the baroclinic vorticity is thus triggered gradually over the bubble interface. Interestingly, compared to the downstream end of a light gas interface, there is less growth of rolled-up vortices downstream of a heavy gas interface.

Fig. 13 illustrates the Atwood number impact on the vorticity contour in the shock-driven pentagonal interface at different time instants. Significant variations in vorticity generation are noted for the various Atwood numbers following the interaction. When the Atwood number is negative ($At < 0$), as illustrated in Figs. 13(a)–(b), the positive vorticity is generated on the upper two sides of the pentagonal and the negative vorticity is generated on the lower two sides of the interface. This is a result of the IS wave moving along the pentagonal interface in a left-to-right direction. Consequently, the pressure gradient is across the upstream IS wave and the density gradient is distributed radially outward at the bubble boundary. Moreover, the upper part of the pentagonal interface has a vortical structure with positive vorticity in the center and tails of negative vorticity surrounding it, whereas the lower part displays the opposite pattern. When $At \approx 0$ i.e. $At = 0.074$, as shown in Fig. 13(c), the leftmost and rightmost vertical surfaces of the pentagonal exhibit no vorticity generation. Interestingly, when Atwood number is positive i.e. $At > 0$, as illustrated in Figs. 13(d)–(e), a significant quantity of negative and positive vorticity are generated on the upper and lower parts of the pentagonal interface. At the corners of pentagonal, there is a significant quantity of vorticity in both the upper and lower vortex pairs (UVR and DVR). Additionally, following shock focusing, the deformed shock wave generates a tiny amount of both positive and negative vorticity on the upper and lower interfaces. The dominant positive (negative) vorticity is also gathered on the upper-half (lower-half) plane in the inward jet head, specifically at $At = 0.466$ and $At = 0.667$. The higher amplitude attenuation of the deformed shock wave upon impacting the pentagonal's downstream pole could be the cause of the dominating positive (negative) vorticity that is accumulating in the inward jet head on the upper-half (lower-half) plane. The relationship between the vorticity deposition and the jet formation is implied by the inward jet movement that is encouraged by the positive (negative) vorticity on the upper (lower) jet head.

4.5. Integral diagnostic

In this section, we investigate the Atwood number impact on the spatial integrated quantities related with the flow fields. Firstly, the production terms of the VTE are explored in depth, which are defined in the subsequent. The spatial integrated field of dilatational vorticity production is computed as follows:

$$\omega_{dil}(\tau) = - \frac{\int_D |\omega(\nabla \cdot \mathbf{u})| dx dy}{\int_D dx dy}, \quad (28)$$

where D represents the entire computational domain. While, the spatial integrated field of baroclinic vorticity production is given by

$$\omega_{bar}(\tau) = \frac{\int_D \left| \frac{1}{\rho^2} (\nabla \rho \times \nabla p) \right| dx dy}{\int_D dx dy}. \quad (29)$$

Fig. 14 illustrates the Atwood number impact on the dilatational ($\omega_{dil}(\tau)$) and baroclinic ($\omega_{bar}(\tau)$) vorticity production terms in the shock-driven pentagonal interface. The plots demonstrate how, during the interaction, both terms acquire considerable values and quickly increase after traversing the shock wave. Due to compressibility effects resulting from local regions of compression and expansion, the plot of the dilatational vorticity production term demonstrates the presence of locally stretched structures surrounding the vortex core, as illustrated in Fig. 14(a). Interestingly, the dilatational vorticity term has an enhanced value at $At = 0.074$, although its modest magnitude value is noticed at the maximum negative and positive Atwood numbers, namely $At = -0.773$ and $At = 0.667$. On the other hand, the misalignment of pressure and density gradients is captured in the plot of the baroclinic vorticity production term, which in turn displays the reflected shock structures and vorticity created by the presence of contact discontinuity, as shown in Fig. 14(b). With a value of $At = 0.074$, the pentagonal interface has the smallest baroclinic vorticity, while $At = -0.773$ has the biggest value, which implies that the ambient gas is increasingly entrained into the distorted pentagonal interface.

We now examine two additional integral quantities – enstrophy (Ω) and kinetic energy (K.E.) – to further study the Atwood number impact on physical phenomena of vorticity generation. The area integral of the vorticity square in the flow field represents the time evolution of the enstrophy, which is defined as

$$\Omega(\tau) = \int_D \omega^2 dx dy. \quad (30)$$

The spatial integral of the square of the velocity in the flow field represents the time development of the kinetic energy (K.E.), which is defined as

$$\text{K.E.}(\tau) = \frac{1}{2} \frac{\int_D \mathbf{u}^2 dx dy}{\int_D dx dy}. \quad (31)$$

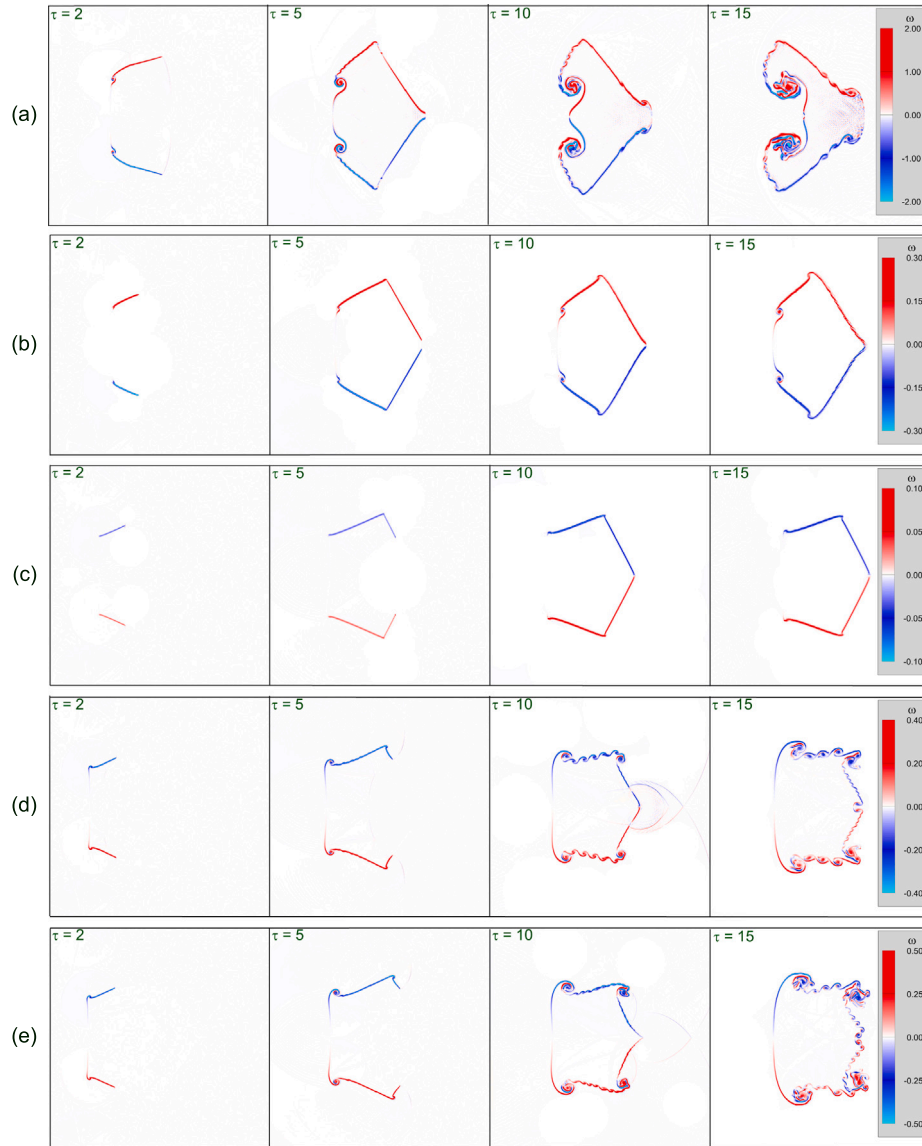


Fig. 13. Atwood number impact on the vorticity contour in the shock-driven pentagonal interface at different instants: (a) $At = -0.773$, (b) $At = -0.218$, (c) $At = 0.074$, (d) $At = 0.466$, and (e) $At = 0.667$.

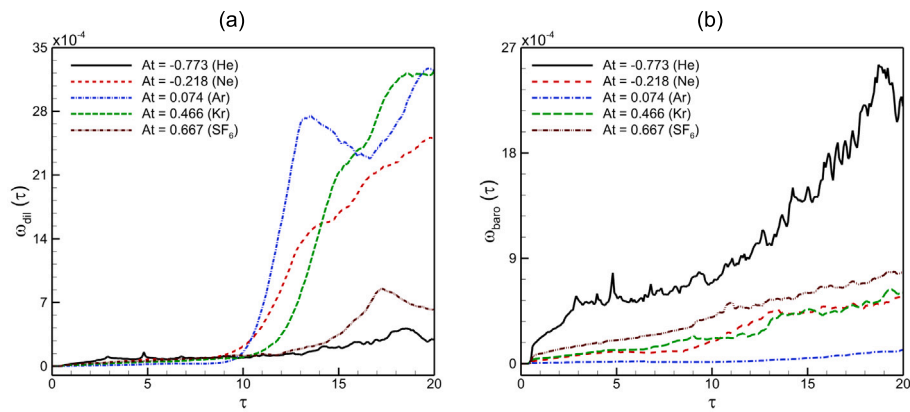


Fig. 14. Atwood number impact on spatial integrated field of (a) dilatational vorticity production term, $\omega_{dil}(\tau)$ and (b) baroclinic vorticity production term, $\omega_{bar}(\tau)$ in the shock-driven pentagonal interface.

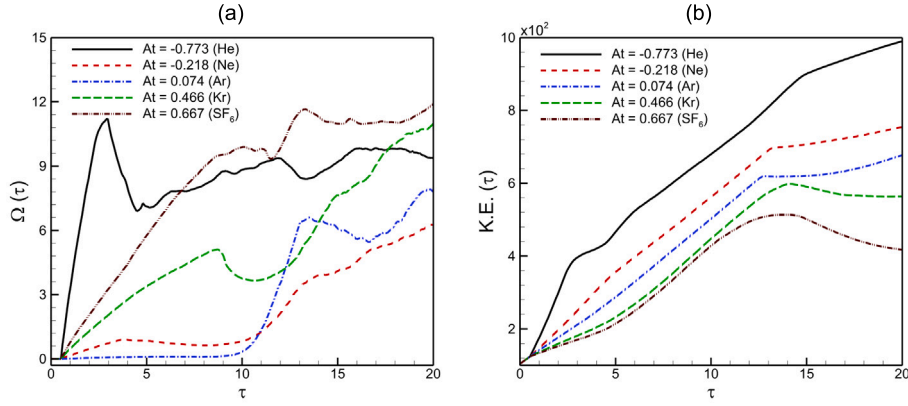


Fig. 15. Atwood number impact on spatial integrated field of (a) enstrophy and (b) kinetic energy in the shock-driven pentagonal interface.

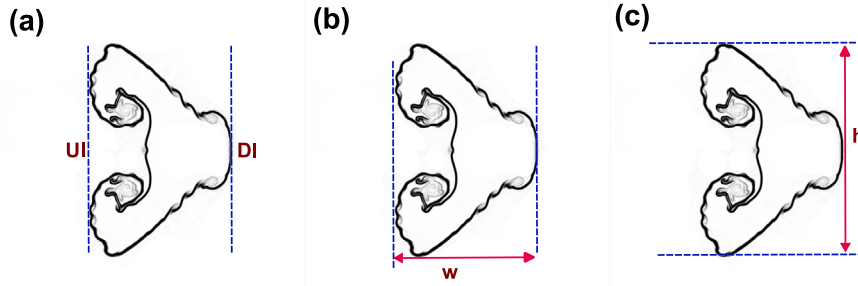


Fig. 16. A diagrammatic representation of the deformation parameters in the shock-driven pentagonal interface: (a) upstream and downstream interface (UI, DI) points, (b) interface width (w), and (c) interface height (h).

In the interaction between the IS wave and the pentagonal interface, Fig. 15 illustrates the Atwood number impact on the spatial integrated fields of enstrophy and kinetic energy in the shock-driven pentagonal interface. The enstrophy is zero until the shock wave reaches the upstream pole of the bubble. Baroclinic vorticity production leads to an increase during the shock wave passage. When the IS and RS waves impinge on the bubble interface, the enstrophy increases. Subsequently, the enhanced vorticities promote the mixing of gases inside and outside the gas bubble and thus accelerate the transfer and consumption of energy, which can weaken the enstrophy intensity in the bubble region gradually, as shown in Fig. 15(a). The same phenomenon is observed in all Atwood numbers. Only overall enstrophy levels differ as Atwood number value increases or decreases. The evolution of the kinetic energy is plotted in Fig. 15(b), which is different and dependent upon Atwood numbers. It is obvious that the kinetic energy is highly raised with the decreasing Atwood numbers.

4.6. Interface features

To shed light on the Atwood number impact, we present a quantitative study of the interface features of shock-driven pentagonal interfaces in this section. As shown in Fig. 16, the interface features of the pentagonal are identified by the displacement of the upstream interface (UI), downstream interface (DI), interface width (w) in the x -direction, and interface height (h) in the y -direction.

Fig. 17 displays the temporal variations of the normalized upstream distance (α) and normalized downstream distance (β) of the computed pentagonal interface with varied Atwood numbers after impinging by the IS wave. These normalized distances are computed as $\alpha = UI/d_{UI}$, and $\beta = DI/d_{DI}$, respectively, where d_{UI} and d_{DI} are chosen as the original distances of UI and DI positions from the IS wave, respectively. Fig. 17(a) shows that α behaves similarly at early stages for all Atwood numbers in the shocked-pentagonal interface. It then accelerates, which

is most likely the result of rarefaction waves that are produced when the shock wave interacts with the downstream contact. Strong compression effects cause the α value of light pentagonal interface with $At < 0$ to exceed that of heavy pentagonal interface with $At > 0$. One can be observed that the displacement of α becomes smaller with increasing the Atwood number, as the velocity of the upstream interface increases. Conversely, during the initial phase, the β value of pentagonal bubbles remains motionless until it encounters the secondary transmitted shock wave, as illustrated in Fig. 17(b). Once the STS wave has cleared, β attains a certain velocity that triggers the acceleration of the vortex rings (UVR, DVR). The velocity of β , on the other hand, increases when the Atwood number decreases, but the displacement of β increases.

Fig. 18 displays the temporal variations of the normalized interface width (w^*) and interface height (h^*) for the computed pentagonal interfaces with different Atwood numbers. These normalized quantities are defined as $w^* = (w - w_0)/w_0$, and $h^* = (h - h_0)/h_0$, respectively, with subscript '0' denoting the initial value. During the early phase of interaction, the IS wave compression drastically shortens the width of the evolving interface for all At values when it strikes the pentagonal, as shown in Fig. 18(a). After the compression phase, the interface length is almost constant for a long evolving time. In case of $At = 0.667$ and $At = 0.466$, the interface length decreases until $\tau = 8$ and $\tau = 12$, respectively. After then, it increases slowly due to the growing the vortex pairs (UVR, DVR), as it can also be seen from Figs. 9–10. It is interesting to note that the vortex pair's continuous spin causes the interface height to rise until reflected shock waves strike the interface and reduce the rate of interface height growth. The maximum interface height is observed in the case of $At = -0.773$, while the minimum interface height is witnessed for $At = 0.074$, as seen from Fig. 18(b).

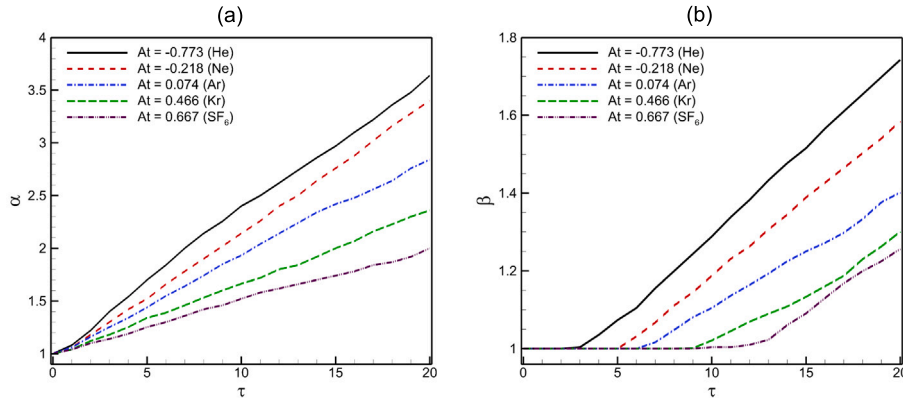


Fig. 17. Atwood number impact on the time variations of (a) normalized upstream interface (α) and (b) normalized downstream interface (β) in the shock-driven pentagonal interface.

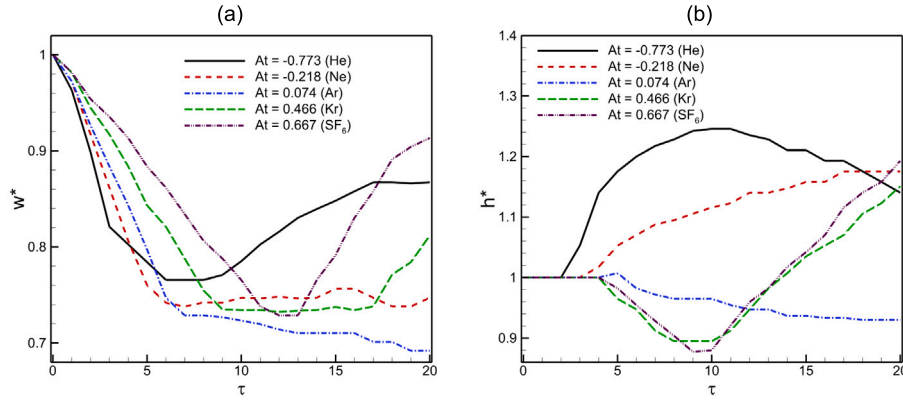


Fig. 18. Atwood number impact on time variations of (a) normalized interface width (w^*), and (b) normalized interface height (h^*) in the shocked-pentagonal interface.

4.7. Comparative flow field evolution between shocked-square and pentagonal interfaces

Finally, in this section, a comparative study of the flow field evolution between the shocked-square and pentagonal interfaces is presented. For this purpose, two different Atwood numbers— $At = 0.667$ and $At = -0.773$ are selected, according to filled interface gases (SF_6 , and He). While, N_2 gas is assumed as the ambient gas of the interfaces. All numerical simulations are done with the same shock wave strength $M_s = 1.2$.

Fig. 19 depicts a series of the numerical Schlieren images for flow field evolution of the shocked-square and pentagonal He interfaces in N_2 gas at $At = -0.773$. It has been demonstrated that the propagation speed of the IS wave is higher than that in surrounding gas due to the small acoustic impedance in both square and pentagonal interfaces. The resulting diverging forms of the interfaces are caused by the generated DTS wave inside the bubble moving faster behind the IS wave. As illustrated in Figs. 19(a)–(b), a distinct flow field evolution from the pentagonal interface is noted in the shocked-square He square interface. For instance, two triple points (TP1, TP2) are seen in the square interface while one triple point (TP) is visible in the pentagonal interface during the irregular refraction mechanism ($\tau = 2$). Interestingly, no downstream vortex rings (DVR), third transmission shock (TTS), or reflected rarefaction shock (RRS) wave are detected in the shocked-square He interface. Moreover, the pentagonal interface contains a distinct re-entrant He jet head from the square interface. In contrast to the square interface, a thicker connecting bridge (CB) is seen in the pentagonal interface ($\tau = 20$).

Fig. 20 illustrates a series of the numerical Schlieren images for flow field evolution of the shocked-square and pentagonal SF_6 interfaces in

N_2 gas at $At = 0.667$. In both interfaces, the propagation speed of the IS wave is smaller than that in surrounding gas due to the large acoustic impedance. As a result, the interfaces take on a convergent structure and the generated DTS wave inside the interface moves slower behind the IS wave. A unique flow field evolution from the pentagonal interface is observed in the shocked-square SF_6 square interface, as shown in Figs. 20(a)–(b). For example, with pentagonal interface compared to square interface, a longer size of lateral transmitted shock (LTS) wave is observed. Larger upstream vortex rings (UVR) and smaller downstream vortex rings (DVR) are detected at square interface, but these vortex rings are observed in pentagonal interfaces in opposite patterns. It is noteworthy that in the shocked-square interface, an outward jet is produced in the center-line of downstream, whereas in the shocked-pentagonal interface, an inward jet is produced.

5. Concluding remarks and outlook

In fluid mechanics, the Atwood number plays a critical role to understanding the flow physics behind the evolving hydrodynamic instability. This study investigates numerically the Atwood number impact on the shock-driven hydrodynamic instability at a backward-facing pentagonal interface in its early stages with shock Mach number $M_s = 1.2$. The shock-driven pentagonal interface has a unique property in which the incident angle of the shock wave is constant along the interface edge. This property provides the ideal circumstances for the shock-refraction mechanism. To examine the Atwood number effect, we consider five distinct gases, including sulfur hexafluoride, krypton, argon, neon, and helium, which are filled inside the pentagonal interface and surrounded by the ambient N_2 gas. An in-house modal discontinuous Galerkin method is employed to solve two-dimensional

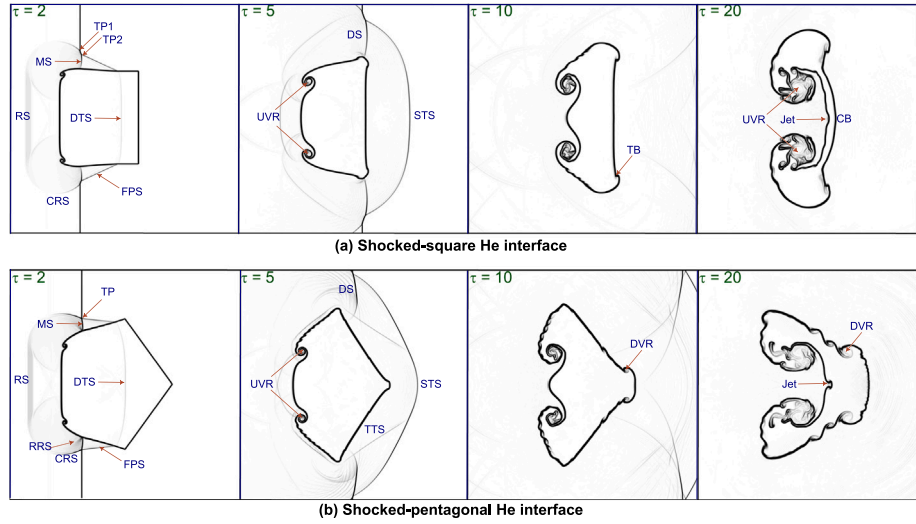


Fig. 19. Comparative flow field evolution between the shocked-square and pentagonal He interfaces with $At = -0.773$.

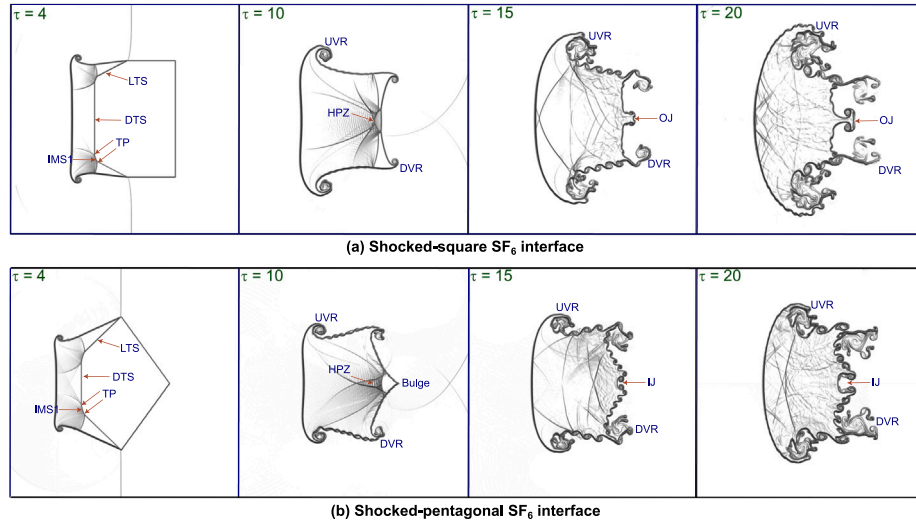


Fig. 20. Comparative flow field evolution between the shocked-square and pentagonal SF_6 interfaces with $At = 0.667$.

NSF equations for two-component gas flows for numerical simulations. The solver resolved well the vortices originating from the pentagonal interface, and other intricate wave patterns. The numerical results for the validation study are compared with the existing experimental results, and they show good agreement.

The simulation results show that the growth of shock-driven hydrodynamic instability at the pentagonal contact is significantly influenced by the Atwood numbers. Flow field evolution is significantly altered by the effects of the Atwood number, resulting in complex wave patterns, the production of vortices, the generation of vorticity, and interface deformation. To be more precise:

- For $At < 0$, the transmitted shock wave travels faster than the incident shock wave due to the low density of pentagonal interface, and two pairs of vortex rings (UVR, DVR) at the pentagonal corners are generated, which develops with time and dominates the flow fields. Furthermore, three distinct types of reflection-transmission processes are seen.
- For $At \approx 0$, the IS and DTS waves travel with similar speeds. Once the DTS wave reaches the rear surface of the pentagonal, a shock wave is transmitted into the ambient, while, RTS wave traveling upstream is generated inside the pentagonal. Besides it, two pair

of tinny vortex rings (UVR, DVR) at the upstream/downstream corners and an inward jet are observed.

- For $At > 0$, because of the substantial density of the pentagonal interface, the IS wave travels faster than the DTS and LTS waves. A decrease in the Atwood number is found to also accelerate the shock-focusing time and a different type of inward jet is generated. Remarkably, vortex pairs originate from both the upstream and downstream corners, with the upstream vortex pair eventually controlling the flow shape.
- For $At = -0.773$, the pentagonal experiences the fastest shock propagation, whereas it experiences the slowest for $At = 0.667$.
- Considerable amounts of both positive and negative vorticity are produced on the bubble's top and lower interfaces for $At < 0$, but a reversal pattern of same mechanism is seen for $At > 0$.
- The time-variations of the integral quantities and interface features are examined from a quantitative analysis perspective. Interestingly, the interface height is lowest at $At = 0.074$ and largest at $At = -0.773$.
- Based on the numerical Schlieren images, a comparative examination of the flow field evolution between the shocked-square and pentagonal surfaces is presented. There are noticeable differences in flow fields between the square and pentagonal interfaces,

specifically in the form of wave patterns, vortex rings, and jets formation.

The main objective of this study was to explore the Atwood number impact on the shock-driven pentagonal interface. Interestingly, the orientation of the pentagonal interface significantly impacts the behavior and characteristics of RMI, influencing factors such as instability growth, vortex dynamics, mixing efficiency, and overall flow evolution. For example, the specific angles and sides of the pentagon relative to the shock direction can influence the formation and dynamics of vortices. This affects how efficiently the kinetic energy from the shock is converted into rotational motion, impacting the subsequent mixing and turbulence. Therefore, in this context, the numerical introspection based on high-order DG solver will be reported to analyze the orientation impact of shock-driven pentagonal interface in the future.

CRedit authorship contribution statement

Sham Bansal: Writing – original draft, Visualization, Investigation. **Ashok Kumar:** Writing – original draft, Visualization, Investigation, Formal analysis. **Akshay Saini:** Visualization, Investigation, Formal analysis. **Anup Singh Negi:** Writing – original draft, Investigation. **Satyvir Singh:** Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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